Response to Reviewers, Round 2

Inferring circadian phases and quantifying biological desynchrony across single-cell transcriptomes

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We thank both reviewers for their second round of comments. Below we reproduce each comment (in *italics*) followed by our response.

A note on numbering. Figure, panel and equation references in our responses are to the revised manuscript unless stated otherwise. References of the form Fig. Reb N point to figures of this document or of the first round response.

Summary of main changes.

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Reviewer 1

R1.1 — Phase at which the Twin-Dataset is constructed

The phase at which the Twin-Dataset is constructed is never stated. The Methods define the twin as a synthetic dataset reproducing the per-cell library sizes and sequencing depth of the data under study, in which every cell shares the same circadian phase, and Eq. (15) then subtracts the resulting $\sigma_{technical}$ at "the group's sequencing depth". Which phase the cells are placed at is not specified anywhere. This is not a detail: the paper itself shows $\sigma_{technical}$ varying roughly two-fold around the cycle (Fig. S1F), so the choice materially determines every reported σ_{bio}. The response to my earlier point makes clear that the twin sits at the phase of the group it mirrors, and that this is precisely what protects the estimator from the phase-dependence concern I raised — but a reader of the manuscript alone cannot reconstruct the central estimator. One sentence in the Twin-Dataset paragraph would close this.

Response.

R1.2 — Derivation of the period spread

Two problems in the derivation of the period spread. The Methods state that fixed period heterogeneity and within-cycle stochastic phase fluctuations "both produce a linearly growing cSTD, so they cannot be separated from a slope alone". Diffusive phase accumulation gives a variance linear in time and therefore a cSTD growing as the square root of time, whereas period heterogeneity gives a cSTD linear

in time; the two have different time dependence. The two references cited in that same sentence treat the contributions as distinguishable in principle. Please clarify.

Response.

R1.3 — Explanation of the phase dependence of $\sigma_{\text{technical}}$

The explanation offered for the phase-dependence of $\sigma_{technical}$ does not match the periodicity reported. The Results attribute the variation of $\sigma_{technical}$ across the cycle to "the non-uniform distribution of peak phases across the rhythmic gene set", which is a once-per-cycle effect. Yet the SABER-FISH paragraph describes the modulation recovered in real data as 12h-periodic, and Fig. S1F shows two maxima per cycle. The dominant mechanism — that a small phase error perturbs predicted expression in proportion to the local steepness of the cosine, which is flat at peak and at trough and steepest between them, giving two maxima per cycle — is stated clearly in a footnote of the response and appears nowhere in the manuscript. Uneven peak phases account only for the weaker once-per-cycle component. Please move the correct explanation into the Results; it is two sentences, and it matters because this modulation is what makes the per-time-point σ_{bio} values uneven, which the same paragraph goes on to discuss.

Response.

R1.4 — Supplementary tables and reproducibility

Supplementary tables and reproducibility. Tables S1, S2 and S3 are cited in the main text — the Extended-Set gene parameters, the per-cell-type and per-time-point variance components, and the Drosophila-Set parameters respectively — but are not present in the supplementary file supplied to me. Please confirm they ship with the paper rather than only with the code repository; Table S2 in particular is the item I wanted to see and is the one a reader will want alongside Fig. 3J. On the code itself, the position is now satisfactory in substance: both repositories are public and the preprocessed data are deposited. I can think of one potentially useful addition, which is the lightweight cell bootstrap that the authors describe as running in seconds from the deposited per-cell phases. I think this would be a very useful artefact in the repository if shipped as a standalone script.

Response.

Reviewer 2

R2.1

In response to my previous comment on the validity of their variance decomposition (R2.2 in the author's response), the authors argue that the conditioning variable S in their Equation 13 (revised main text) is one replicate at one ZT time point and that the $\sigma_{bio}/\sigma_{tech}$ split is a "further decomposition that lives entirely inside the within term". While this is OK, the author's decomposition purely within a ZT still must obey the law of total variance, since it is an identity that holds for any conditioning variable. Therefore, if Eq. 14 is to be more than an analogy, it must itself be a total-variance decomposition within one single ZT timepoint with respect to some conditioning variable, and the authors need to precisely define what that variable is. Otherwise, Eq. 14 appears quite ad hoc and it is impossible to interpret what the terms σ_{bio} and σ_{tech} are supposed to represent in a statistically rigorous framework.

Response. We thank the reviewer. The reviewer is right: a split of the variance inside one ZT has to follow the law of total variance (LTV), so we must say which variable we condition on. Writing this out showed us where Eq. 14 of the main text is exact and where it is only approximate, and it led us to an additional estimator that avoids the approximations.

The conditioning variable is the true phase θ of the cell.

In short: the conditioning variable is the latent true phase θ of a cell in one replicate at one ZT. σ^2_{bio} is the variance of θ across those cells, and σ^2_{tech} is the average, over those cells, of the variance of the inferred phase at fixed θ. In the reviewer's terms, S (one replicate at one ZT) is the outer split, and θ is the variable inside the within S term, which is the further decomposition we referred to.

In detail, take one group b of n_b cells (one replicate at one ZT). Each cell is generated in two steps. Its true phase is drawn, $\theta \sim p_b(\theta)$, with mean μ_b and standard deviation $\sigma_{\text{bio},b}$. This is the biological variability, and θ is never observed. Then its inferred phase is drawn, $\hat{\theta} \sim p(\hat{\theta} \mid \theta)$. This distribution comes from the count noise and from the inference, and it is the technical part. Write $m(\theta) = \mathbb{E}[\hat{\theta} \mid \theta]$, $\varepsilon(\theta) = m(\theta) - \theta$ for the bias of the estimator, and $f(\theta) = \text{Var}(\hat{\theta} \mid \theta)$. The LTV with θ as the conditioning variable reads

$$\underbrace{\text{Var}(\hat{\theta})}_{V_b} = \underbrace{\mathbb{E}_{\theta\sim p_b}\big[\,\text{Var}(\hat{\theta} \mid \theta)\big]}_{T_b\ \text{(technical)}} + \underbrace{\text{Var}_{\theta\sim p_b}\big(\mathbb{E}[\hat{\theta} \mid \theta]\big)}_{B_b\ \text{(biological)}}, \tag{1}$$

so $T_b = \int p_b(\theta)\, f(\theta)\, d\theta$ and $B_b = \text{Var}_{\theta\sim p_b}\big(m(\theta)\big)$. This holds for every group and every shape of p_b. If the estimator has no bias, $m(\theta) = \theta$, then $B_b = \sigma^2_{\text{bio},b}$. In practice the bias is not zero (for sparse cells, estimates are pulled toward the trough of the aggregate circadian counts, about ZT21), so $B_b \neq \sigma^2_{\text{bio},b}$ in general, and this matters below. The outer average runs over the cells of one group only, so it does not cover the whole cycle. Averaging across groups is a separate step, treated in R2.3.

The circular form.

Phase wraps around at 24 h, so from here on we work on the circle. The variance of a group is $V_b = -2\ln \bar{R}_b$, where $\bar{R}_b$ is the length of the mean of the unit vectors $e^{\mathrm{i}\hat{\theta}}$ over the n_b cells, and $f(\theta) = -2\ln r(\theta)$ (defined below). Define $\rho(\theta) = \mathbb{E}[e^{\mathrm{i}\hat{\theta}} \mid \theta] = r(\theta)\, e^{\mathrm{i}m(\theta)}$, so that $m(\theta)$ is now the

mean direction. Conditioning on θ gives the exact identity

$$\mathbb{E}\big[e^{\mathrm{i}\hat{\theta}}\big] = \mathbb{E}_{\theta\sim p_b}\big[\rho(\theta)\big]. \tag{2}$$

If $r(\theta) = r$ is constant over the group and $m(\theta) = \theta$, the length of the right side is r times the length of $\mathbb{E}_{\theta\sim p_b}[e^{\mathrm{i}\theta}]$. Taking $-2\ln$ then gives $V_b = f + \sigma^2_{\mathrm{bio},b}$, with $f = -2\ln r$ and $\sigma^2_{\mathrm{bio},b} = -2\ln|\mathbb{E}_{\theta\sim p_b}e^{\mathrm{i}\theta}|$ the circular variance of θ. This is Eq. 14, and it is exact under these two conditions.

Where our earlier estimator sits.

The observed V_b is known, while T_b and B_b are not. Our earlier estimator (Eq. 14 of the main text) fills in Eq. (1) as follows. For the biological term it takes $B_b \approx \sigma^2_{\mathrm{bio},b}$, which is exact if $m(\theta) = \theta$. For the technical term it simulates a Twin dataset: a copy of the group in which all cells have the same true phase (no biological spread), the genome wide library size (UMI count per cell) of the group, and the fitted model. The variance of the inferred phases of the Twin is f at that one phase. So $V_b \approx f(\mu_b) + \sigma^2_{\mathrm{bio},b}$. This is Eq. (1) with two approximations: (A1) T_b is evaluated at a single phase instead of averaged over p_b; (A2) the bias is neglected in B_b. For a small bias and a narrow p_b, the error $\Delta_b = V_b - \big(f(\mu_b) + \sigma^2_{\mathrm{bio},b}\big)$ has one part for each approximation.

- A1, technical variance read at one phase. The cells of the group sit at different true phases, and the technical variance f changes with the phase. The exact technical term is the average of f over these phases, while the Twin reads f at the center only. The estimate of $\sigma^2_{\mathrm{bio},b}$ is the data variance minus the Twin variance. Where f curves upward around μ_b, the average is larger than the value at the center, so the Twin variance is too small and the estimate of σ_{bio} too large. Where f curves downward, it is the other way. The error grows with $\sigma^2_{\mathrm{bio},b}$ and with how strongly f curves along the phase. Because phases are averaged as unit vectors on the circle, a smaller extra term appears, which is never positive.

- A2, bias left out of the biological term. Inferred phases are pulled toward the trough of the aggregate circadian counts (about ZT21). If the pull changes across the phases of the group, the inferred phases are squeezed together or spread apart compared with the true ones, and B_b is smaller or larger than $\sigma^2_{\mathrm{bio},b}$. Cells on both sides of the trough move toward it and end up closer together, so near the trough the estimate is too small. This error also grows with $\sigma^2_{\mathrm{bio},b}$.

So Eq. 14 of the main text is approximate within one ZT, and these are the reasons. Averaging over groups (Eq. 16 of the main text) can cancel part of this error. We come back to this below.

The new estimator: the vector form.

Instead of correcting these terms one by one, we start from Eq. (2) itself, which is exact. Its left side comes from the data: the mean $\bar{z}_b$ of the unit vectors $e^{\mathrm{i}\hat{\theta}_c}$ of the cells, with $\bar{R}_b = |\bar{z}_b|$. On the right, $\rho(\theta)$ is the technical part and p_b is the biological part. We measure ρ with Twin cells on a grid of 24 phases, write it as a Fourier series, and average it over a wrapped normal p_b of width σ. This gives the expected $\bar{R}^2_b$ of a group as a known function of σ, and $\hat{\sigma}_{\mathrm{bio},b}$ is the value of σ that matches the data. A1 and A2 disappear, because ρ is averaged over the whole of p_b and its direction holds the bias, and no variances are added. Appendix A gives the steps, what is removed and the assumptions that remain.

The vector form in simulation.

We simulated 15 clock genes and 6 groups of 1000 cells at phases 4 h apart, at 3k, 10k and 30k UMI per cell, with a true spread of zero (all cells of a group at the same phase), 1.0 h or about 2.5

h (von Mises), and 5 replicates of the whole experiment. The data are generated from a perturbed copy of the fitted gene profiles, so the fitted model is not the generating model. The truth of a group is the circular standard deviation of the true phases of its cells. Fig. Reb 1A shows the absolute error of each group for the Twin method and the vector form. Fig. Reb 1B compares the mean error of a single group with the mean error of the single value over the 6 groups of one replicate (Eq. 16 of the main text), $\sqrt{\sum_b n_b \hat{\sigma}^2_{\text{bio},b} / \sum_b n_b}$, whose truth is the same average taken over the true values. When the estimate of a group falls below the technical value, it is counted as zero for both methods, in both panels.

For a single group the vector form is more accurate. With a true spread of about 2.5 h, its mean absolute error is 0.46 h against 0.60 h for the Twin method at 3k, 0.17 against 0.24 h at 10k, and 0.07 against 0.15 h at 30k. With a spread of 1.0 h the difference is smaller (0.51 against 0.62 h, 0.26 against 0.28 h, and 0.14 against 0.16 h). For both methods the error falls as the depth goes up.

The single value over the 6 groups has a mean error 1.6 to 6 times smaller than a single group (Fig. Reb 1B). At 10k and 30k it is 0.08 h or less for both methods and both spreads, and the two methods give the same result. At 3k the vector form is better (0.24 against 0.35 h for a spread of 2.5 h, and 0.26 against 0.39 h for 1.0 h).

The figure does not show the case with no spread. There 43 to 67% of the groups, depending on depth and method, fall below the technical value and get an estimate of zero, and the others get a small positive value. The single value over the groups therefore cannot go below zero, and its mean error is about 0.4 h at 10k and 30k for both methods (0.7 to 0.8 h at 3k). At these depths an estimate of σ_{bio} below about 0.4 h cannot be told apart from zero. The same control, with a σ_{bio} that changes across time points, is in R2.3c (Fig. Reb 4B).

The oracle used in R2.3c is a reference, not an estimator: it re-simulates each group at the true phases of its cells with the fitted model, so it has no A1 or A2 error.

The average over groups.

The decomposition is defined and estimated for each group b (one replicate at one ZT). The single value over groups, Eq. 16 of the main text, $\sqrt{\sum_b n_b \hat{\sigma}^2_{\text{bio},b} / \sum_b n_b}$, is an extra step and not part of the definition. It estimates one σ_{bio} only if all groups share one. This is not the case in the SABER-FISH experiment, where σ_{bio} rises with time, so there we do not use it and read the estimates per time point (R2.3c). When σ_{bio} changes across groups, the single value is their average weighted by the number of cells, and it hides the change (R2.3c). Our first round statement that σ_{bio} is "a property of a population sampled across the cycle, not of an individual ZT" was too strong.

We do not need the average to get a usable estimate: one group is enough from about 10k UMI per cell, with an error of 0.24 h for the Twin method and 0.17 h for the vector form at a spread of 2.5 h (Fig. Reb 1A). When σ_{bio} is the same in all groups around the clock, the average reduces the error on σ_{bio}, by a factor of 1.6 to 6 in the simulation above (Fig. Reb 1B). Part of this gain is the plain averaging of the noise of single groups. The other part is that it cancels part of the Twin error Δ_b. Take time points evenly spaced over the cycle, with the same $\sigma_{\text{bio},b}$, depth and number of cells in every group. The main parts of A1 and A2 change sign around the cycle: f curves upward at some phases and downward at others, and the cells are squeezed at some phases and spread apart at others. Over a whole cycle these parts average to zero, so they cancel. With N evenly spaced time points the cancellation is exact except for the parts of f and of the bias that repeat a multiple of N times per day. The other parts do not cancel: the small circle term of A1 keeps one sign, and so does a smaller part of A2 that grows with the square of the squeezing. So a small error is left, and it is included in the error of the single value in the simulation above. With few time points the cancellation is weaker (R2.4).

Which estimator the paper uses.

The decomposition is the same for both methods, namely Eq. (1) with θ as the conditioning variable. The two methods differ only in how they estimate the technical part. The Twin method evaluates it at one phase. The vector form averages it over p_b and keeps the bias inside the prediction. The claims of the paper rest on the single value over the groups (Eq. 16 of the main text) and on slopes over time points. In simulation, from 10k upward, the two methods agree there. The error of the single value is 0.08 h or less for both, and the slopes over time points lead to the same conclusion (R2.3c). On the real data the two methods also agree (Fig. Reb 2). For the mouse cell types of Fig. 3J of the main text the single values differ by 0.02 h or less, except for the Lyve1+ macrophages (3.79 h with the Twin method, 3.88 h with the vector form, less than one standard error apart). For the *Drosophila* clusters of Fig. 4F of the main text the vector form is 0.03 to 0.28 h higher, and σ_{bio} is higher in DD than in LD in the same 3 of 4 clusters. On SABER-FISH the values at single time points are close, but the spread of periods from the slope over time is smaller with the vector form (1.04 against 1.37 h, see R2.3c). Both methods use the same fitted model and the same simulated cells. So their agreement shows that the approximations of the Twin method do not drive these results. It does not test the fit itself, which is the larger source of error (R2.3c). We therefore keep the Twin method as the main estimator, so the results that both reviewers checked in the first round do not change. The vector form is better for a single group and at low depth. At 3k the error of the single value is 0.24 against 0.35 h (spread 2.5 h) and 0.26 against 0.39 h (spread 1.0 h), and for a single group at a spread of 2.5 h it is 0.46 against 0.60 h. We recommend the vector form in those cases. It is in the scRitmo package as an option.

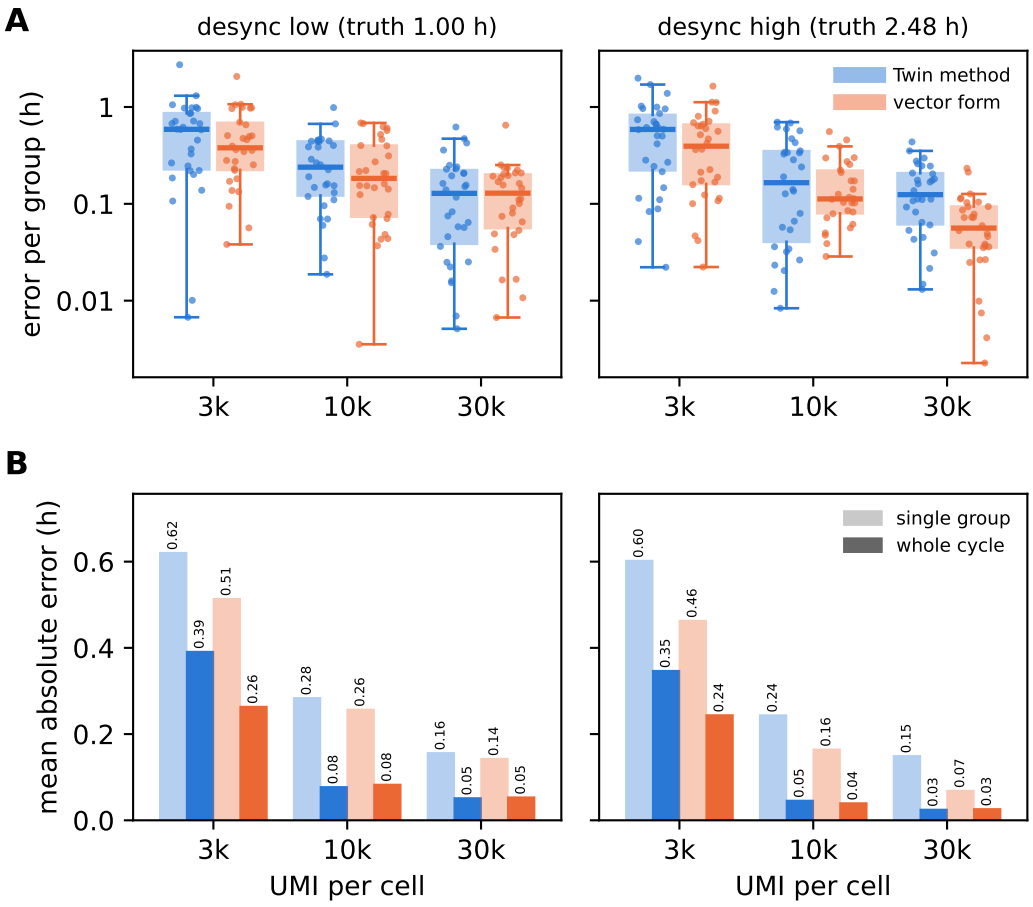


Figure Reb 1: **Absolute error of $\hat{\sigma}_{\mathbf{bio}}$ in simulation, Twin method and vector form.** 15 clock genes, 6 groups of 1000 cells at phases 4 h apart, 5 replicates of the experiment. Columns: low and high desynchrony, with the mean true spread in the panel titles. The x axis is the number of UMI per cell. A group whose estimate falls below the technical value is counted as zero for both methods. **(A)** Error of each group: absolute value of the estimate minus the circular standard deviation of the true phases of the cells of the group, log scale. Boxes: median and quartiles, whiskers to 1.5 times the interquartile range, 30 groups per box. Dots: single groups. **(B)** Mean absolute error of a single group (light bars, 30 groups) and of the single value over the 6 groups of one replicate (dark bars, Eq. 16 of the main text, 5 replicates), for the Twin method (blue) and the vector form (orange). The truth of the single value is the same average taken over the true values of the groups.

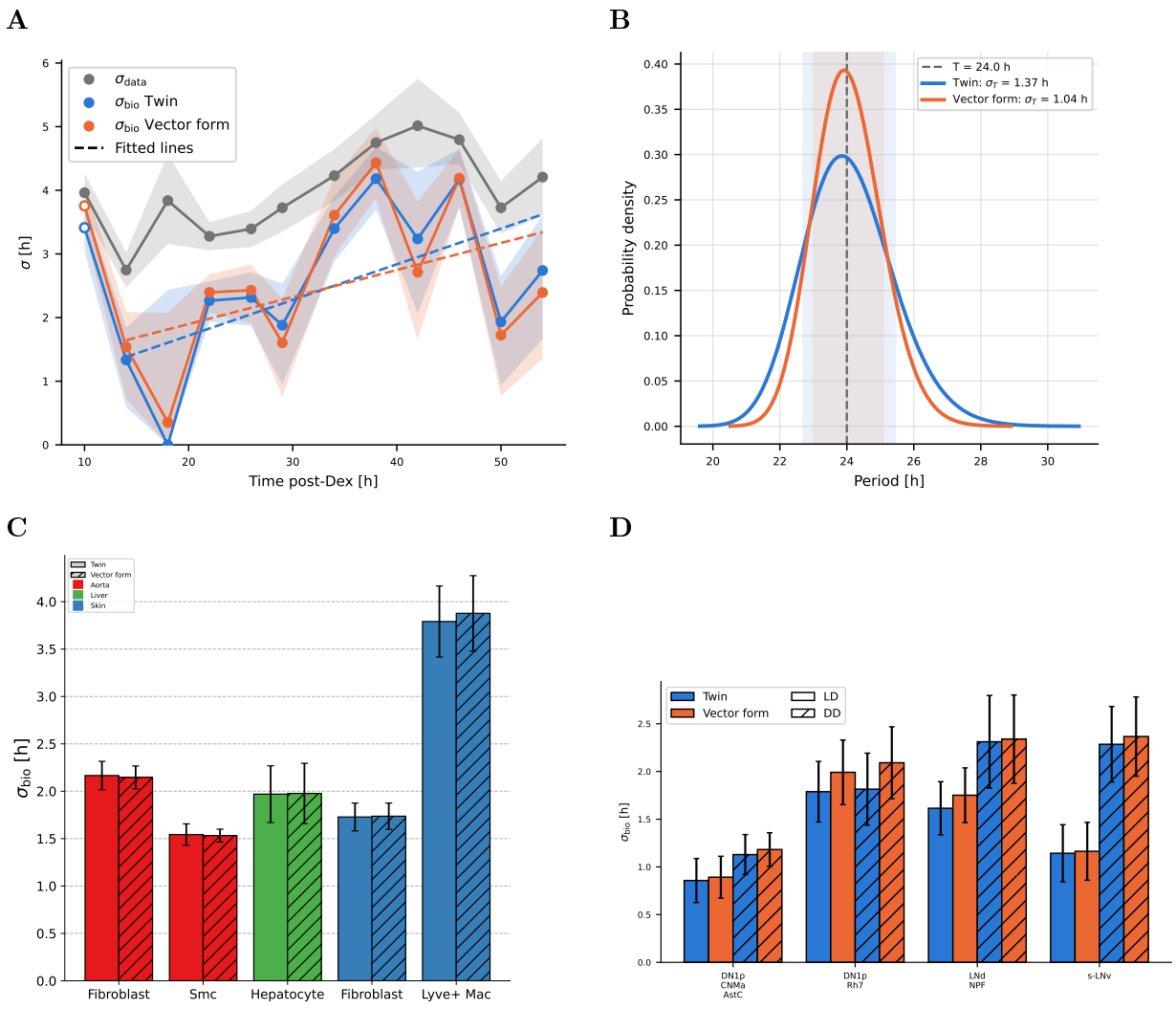


Figure Reb 2: **Twin method and vector form on the real data.** **(A)** SABER-FISH, as in Fig. 2G of the main text. Grey: σ_{data}. Blue: σ_{bio} with the Twin method. Orange: σ_{bio} with the vector form. Bands: 95% intervals of a bootstrap over the cells (2000 draws, technical part held fixed). Dashed lines: lines fitted to the 11 time points after the first one (open markers, left out of the fit), slope not allowed to be negative. **(B)** Distribution of the period obtained from the slopes in A, as in Fig. 2H of the main text. Shaded: mean period plus or minus σ_T. **(C)** Mouse, *Extended-Set*, as in Fig. 3J of the main text. Single value of σ_{bio} over the samples of each cell type (Eq. 16 of the main text). Colors: organ. Plain bars: Twin method. Hatched bars: vector form. Error bars: standard error. **(D)** *Drosophila*, as in Fig. 4F of the main text. Single value of σ_{bio} per cluster. Blue: Twin method. Orange: vector form. Plain bars: light dark cycle (LD). Hatched bars: constant darkness (DD). Error bars: standard error.

R2.2

In the response R2.2, the authors go on to say that they "never define σ_{tech} as the within-term of a latent phase-cluster decomposition (an average over heterogeneous sub-populations)... so there is no averaging over sub-populations with differing variances to reconcile". Does this mean the author's procedure is not consistent with the Law of Total Variance, which must have an averaging procedure

across sub-populations? If so, when the authors say that "At the level of individual time points the additivity is only approximate", is the approximation coming partly from this departure from the Law of Total Variance? These points should be more carefully discussed in the Main Text and Methods, since the current version provides little clarity or discussion on this important issue.

Response. We thank the reviewer. Our first round sentence was wrong and we withdraw it. The reviewer's sub-populations are the cells of one sample grouped by their true phase, which is exactly the conditioning on θ of R2.1. With it the decomposition follows the law of total variance exactly (Eq. (1)). The within term T_b is the average, over these sub-populations, of their technical variance $f(\theta)$, so there is an averaging, as the reviewer said. We had answered at the wrong level, that of the groups S (one replicate at one ZT).

Where the approximation at single time points comes from.

The approximation does not come from a departure from the law of total variance, since the decomposition is exact. It comes in part from how the Twin method estimates the within term: it evaluates f at one phase instead of averaging it over the sub-populations, that is over p_b (A1 in R2.1). The bias (A2) adds to it. This is what we meant in the first round by "partly because σ_{tech} is itself phase dependent", but we did not say so. The paragraph "The average over groups" in R2.1 explains why averaging over groups cancels part of this error and when that average may be used. The vector form (R2.1, Appendix A) does the average over the sub-populations directly.

R2.3

The revised Methods state that Eq. 14 "becomes near-exact ($R^2 \approx 1$) only once variances are aggregated across populations that span the circadian cycle", and draw from this the requirement that "the data must cover most of the circadian cycle". In the response letter, the authors further comment that "σ_{bio} is a property of a population sampled across the cycle, not of an individual ZT". This is made explicit in Eq. 16 of the revised Main Text. This then raises the following important questions which I do not currently understand:

R2.3a

How can this requirement be reconciled with the analysis of the SABER-FISH dataset, where each ZT is analyzed separately to get σ_{bio} and σ_{tech} estimates at each time point? Given that this is the main dataset with direct experimental validation of the σ_{bio} estimates, the analysis of this data and the limitations should be discussed more carefully.

Response. We phrased this badly, and we are sorry that the previous version was not clear. The method does not need an average around the clock. A single time point gives an estimate of σ_{bio}, and the SABER-FISH analysis uses the time points one by one. R2.1 shows the error of single groups next to the error of the average (Fig. Reb 1). The average lowers the error, but only when we have good reason to think that σ_{bio} is the same at all time points.

R2.3b

The aggregation requirements of the method as well as all the simulations the authors have performed (Fig. Reb 3A and 3B) seem to require one ground truth σ_{bio} for the entire population of cells across the different ZT time points. However, this requirement seems directly in conflict with the SABER-FISH data, where σ_{bio} increases with time post Dexamethasone synchronization. Therefore, the

authors need to give a clear justification for applying their method to this dataset. If this is only a very approximate and empirical application of the method, the authors should accordingly tone down their claims of discovering increasing desynchrony and using this dataset as a strong validation of their method.

Response. There is no conflict, as explained in R2.3a. The SABER-FISH analysis does not need the same σ_{bio} at all time points. We show this with simulations in which σ_{bio} rises over time, below in R2.3c.

R2.3c

All simulations in the manuscript appear to hold σ_{bio} to one fixed number across ZT time points, which is why the aggregation procedure seems to work via some cancellations. But if the σ_{bio} per ZT changes, as is the case with the SABER-FISH data, wouldn't this fundamentally create issues in recovering the right σ_{bio} estimates? Aggregation here would probably recover some sort of average of the individual σ_{bio}'s, right? Therefore in order to claim that scRitmo can extract meaningful increase of desynchrony over time, the authors should at least provide simulations where ground truth σ_{bio} increases across ZTs within a single simulation. This simulation should then be analyzed with scRitmo and the results benchmarked for accuracy.

Response. The reviewer is right. When σ_{bio} changes across time points, the single value over all time points (Eq. 16 of the main text) is an average, $\sqrt{\sum_t n_t \sigma_t^2 / \sum_t n_t}$ with n_t the number of cells at time point t, and not the individual values. We ran the simulation the reviewer asks for, in two forms. The first is a simulated copy of the SABER-FISH experiment (Fig. Reb 3). The second is an RNA-seq style experiment at three depths, with several shapes of σ_{bio} (Fig. Reb 4).

SABER-FISH COPY: SIMULATION.

We show both methods because they are two estimators of the same decomposition (R2.1) with different approximations. The Twin method is the one in the paper. The vector form removes the approximations A1 and A2, so if the two agree, those approximations are not what drives the result. We built a copy of the experiment. The data are generated from the fitted model of the real data (template and the negative binomial dispersion of each of the 4 genes), with the 12 real time points and the real number of cells at each. The model uses the cell area where an RNA-seq fit uses the total counts, and we draw the areas of each time point from the real cells of that time point. The true phases of a time point follow a wrapped normal distribution around the phase of that time point, with a known σ_{bio}. We used two profiles. In the first, σ_{bio} rises linearly from 1.0 h at the first time point to 3.5 h at the last. In the second it is constant at 2.24 h, the average of the ramp over the 12 time points, and it checks what happens when there is no increase. The truth at each time point is the circular standard deviation of the true phases of its cells. Each of the 10 replicates of a profile is fitted again with the settings used for the real data (the mean phase of a time point is taken from the data) and then analyzed with the Twin method and the vector form. We also give two oracles as a reference. Both know the true phase of every cell. One simulates each time point again with the fitted model, and the other with the model that generated the data. So the error of the fit is part of what we measure. The true spread of periods σ_T of the ramp (1.40 h) is close to the real Twin value (1.37 h), so the test asks whether a value of this size is recovered, and not whether the real one is right.

SABER-FISH COPY: RESULT.

When the true σ_{bio} rises, the estimates rise with it (Fig. Reb 3A). We fitted a line to the 11

time points after the first one, as for the real data, and converted its slope into the spread of the single cell periods σ_T as in the paper (Fig. Reb 3C). For this the slope is taken as the spread of the angular velocities of the cells, and σ_T is the standard deviation of the periods that follow from them. σ_T is close to 24 h times the slope, so an error of a given size in the slope is an error of the same relative size in σ_T. The true value of the ramp is 1.40 h. The Twin method gives 1.26 ± 0.17 h and the vector form 1.23 ± 0.19 h (mean and standard deviation over the 10 replicates). Both are too small, by 10% and 12%. With all 12 time points they give 1.39 and 1.30 h, which is 1% and 7% too small. When the true σ_{bio} is constant (Fig. Reb 3B), the true value is 0.18 h and not zero, because the line is fitted to the realized spread of the true phases, which varies a little around 2.24 h. The estimates are 0.29 ± 0.14 and 0.20 ± 0.21 h (Fig. Reb 3D). The slope is not allowed to be negative in this fit, as in the paper, so some estimates for the constant profile are exactly zero. The largest value in one replicate of the constant profile, 0.66 h, is about half of the true value of the ramp. We also fitted a line without any constraint on the slope to each replicate and tested whether the slope differs from zero (ordinary least squares, t test on the slope, not shown in the figure). It is significant at the 5% level in all 10 replicates of the ramp and in none of the 10 replicates of the constant profile, for both methods.

At single time points the mean absolute error is 0.58 h (Twin) and 0.57 h (vector form) for the ramp, and the estimates are too high by 0.52 and 0.55 h on average. For the constant profile they are too high by 0.43 and 0.45 h on average. On average over the replicates, single time points are too high by up to 1.2 h in the ramp and 1.3 h in the constant profile (at 29 h after dexamethasone). These bumps repeat across replicates. The oracle with the fitted model shows the same bumps and a similar offset (+0.37 h for the ramp and +0.27 h for the constant profile). The oracle with the model that generated the data does not (-0.04 and -0.08 h). So the Twin method, the vector form and the oracle with the fitted model make similar errors, and the oracle with the generating model does not. This means the error comes from the fit and not from the split of the variance. The vector form does not help here (the mean absolute errors of the two methods differ by 0.01 h) because this error is not the one it corrects. The oracle with the generating model dips to 1.4 h at 18 h in the constant profile, where the time point has the fewest cells (116). We have no other explanation for it. The single value over all time points is 2.89 h (Twin) and 2.90 h (vector form) for the ramp, against a true 2.37 h. This is the average weighted by cell number, and the 2.24 h of the constant profile is the plain average. It lies between the first and the last time point, has the same offset of about 0.5 h, and shows no trend. It is not plotted here.

SABER-FISH COPY: THE REAL DATA.

The real estimates (Fig. Reb 2A, 116 to 362 cells per time point) are far less regular than the simulated ones. The Twin estimate is 3.4 h at 10 h, 1.3 h at 14 h, and zero at 18 h, where the observed spread is smaller than the technical spread. It is about 4.2 h at 38 h and 46 h, and 1.9 h at 50 h. The scatter of the 11 estimates after the first one around a straight line fitted to them (ordinary least squares) is 1.05 h for the Twin method and 1.14 h for the vector form. In the simulation it is 0.50 h and 0.42 h on average, and at most 0.64 h in any of the 20 replicates of the two profiles. We do not know whether the larger scatter is real variation of σ_{bio} between time points or an error that the simulation does not have.

On the real data the two methods agree on an upward trend, but they differ in how clearly it separates from a constant σ_{bio} (Fig. Reb 3E). We use the cell bootstrap of the paper: 2000 draws, the cells of each time point resampled with replacement, the technical part held fixed. With the first time point left out as in the paper, σ_T is 1.37 h for the Twin method (95% interval 0.66 to 1.75) and 1.04 h for the vector form (0.49 to 1.52). With all 12 time points it is 0.85 h (0.34 to 1.15) and 0.53 h (0.20 to 0.91), because the estimate at 10 h is high. The mean of the bootstrap draws

is lower than the value from the real cells (1.23 against 1.37 h for the Twin method, 1.00 against 1.04 h for the vector form), which is why the large markers lie below the crosses in Fig. Reb 3E. The bootstrap does not fit the model again, so these intervals are too narrow. In the simulation a constant profile gave values of up to about 0.65 h for both methods (the largest of 10 replicates, so this is a rough limit, and the oracles reach 0.74 h). With the first time point left out, 98% of the Twin draws and 90% of the vector draws lie above 0.65 h. With all 12 time points, 70% and 22% do. So only the Twin value with the first time point left out is above the range of the constant profile, and only just. The two values from the real cells differ by 0.33 h. In each draw both methods see the same resampled cells, the difference between them is 0.23 h on average, and it changes little from draw to draw (standard deviation 0.06 h). So it comes from the methods and not from the sampling of the cells. The vector form gives higher values at 14 h and 18 h and lower values at 50 h and 54 h (Fig. Reb 2A). At 18 h the Twin estimate is zero, because the subtraction would go below zero. The vector form does not subtract, and it gives 0.35 h there.

The difference between the mean σ_{bio} of the late time points (34 h and after) and of the early ones (22 h and before, 10 h left out) depends less on the method. It is 2.08 h (1.14 to 2.41) for the Twin method and 1.75 h (1.04 to 2.24) for the vector form, and no bootstrap draw is at or below zero for either method.

SABER-FISH COPY: LIMITS.

(i) The true profile is a straight line and has no scatter between time points, while the real one may not. (ii) The simulated time points lie exactly on a 24 h clock. In the real fit the mean inferred phase of a time point is up to 1.7 h away from such a clock, and we did not simulate that. (iii) The generator is the fitted model, so the real data can differ from it in ways the simulation does not cover. (iv) There are 10 replicates, so the standard deviations above are rough. (v) The vector form uses one grid, which is a set of simulated cells at 24 evenly spaced true phases with the cell areas of all time points. The version with one grid per time point gives σ_T of 1.25 and 1.32 h for the ramp (first time point left out, all 12), which are close.

RNA-SEQ STYLE: SIMULATION.

We simulated 15 clock genes and 12 time points of 500 cells, 4 h apart over two days as in the SABER-FISH experiment, at 3k, 10k or 30k UMI per cell (molecules counted over the whole transcriptome). The true σ_{bio} was constant (2.25 h), a ramp from 1.0 h to 3.5 h, or a cosine over the day from 0 to 3 h with its peak at ZT10 (peak of the aggregate circadian counts) or at ZT22 (near the trough). A control had a true σ_{bio} of 0. The cosines are the hard case, because the phase dependent error of the Twin method (R2.1) does not average out when σ_{bio} depends on the phase. We retrained the model on each dataset and analyzed it with the Twin method and the vector form. Both methods estimate the same decomposition, and comparing them checks whether the per group approximations of R2.1 matter when σ_{bio} changes. Numbers below are means over 5 replicates.

RNA-SEQ STYLE: RESULT.

At each time point (Fig. Reb 4A), the estimates follow the ramp and both cosines from 10k UMI upward, and are too high by up to about 0.8 h where the true σ_{bio} is near zero. At 3k the cosine peaking at ZT10 is lost, and the constant profile shows a 24 h pattern that is not in the truth. The single value over all time points (Fig. Reb 4B) is off by at most 0.07 h at 30k and 0.16 h at 10k for the four profiles with spread (single datasets at 10k by up to 0.7 h), and too high by 0.4 to 1.0 h at 3k.

For the ramp, the slope of a line fitted to the 12 estimates, divided by the true slope, is 0.90 (Twin) and 0.93 (vector form) at 30k, 0.85 and 0.91 at 10k, and 0.64 and 0.71 at 3k. The single values

of the two methods lie within the error ranges given above, and the vector form does somewhat better on the slope of the ramp. On the constant and cosine profiles the fitted slope is at most 0.008 h per h in size from 10k upward (0.05 for the ramp), and up to 0.014 at 3k. From 10k upward, a rise of σ_{bio} over time is therefore recovered, about 7 to 15% too small, and a constant or cyclic σ_{bio} does not produce one. At 3k this is not reliable.

RNA-seq style: limits.

With a true σ_{bio} of 0, the single value is too high at low depth: 1.4 to 1.7 h at 3k, 0.5 to 0.8 h at 10k, and -0.1 to 0.4 h at 30k. The vector form sets negative estimates to zero. At 10k and below, a small σ_{bio} is therefore not distinguished from none. In a one day design (ZT 0 to 22), where the ramp and the clock phase are the same axis, the slope ratios are lower (0.77 and 0.86 at 30k, 0.60 and 0.72 at 10k), the constant and cosine profiles give false slopes of up to 0.05 h per h at 10k (the true ramp slope is 0.11), and the vector form is 0.45 h off at 10k for the cosine peaking at ZT10. A rise of σ_{bio} within a single day should be interpreted with care.

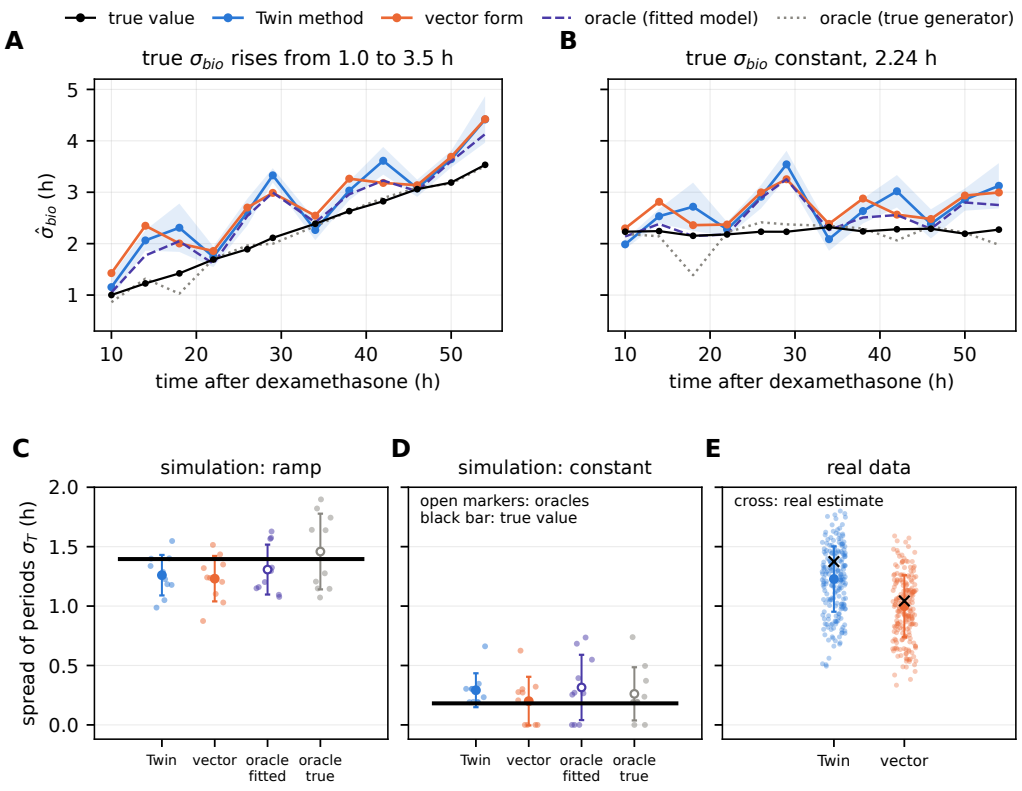


Figure Reb 3: **A simulated copy of the SABER-FISH experiment, with a known σ_{bio} at each time point, next to the real data. (A, B)** Simulation from the fitted model of the real data (4 genes), with the 12 time points and the numbers of cells of the real experiment and the cell areas of the real cells of each time point as library sizes. The model was fitted again on each of the 10 simulated datasets, and the estimates were made with the same settings as for the real data. Estimated σ_{bio} at each time point when the true value rises from 1.0 to 3.5 h (A) and when it is constant at 2.24 h (B). Black: true value, the circular standard deviation of the true phases of the simulated cells of that time point. Blue and orange: Twin method and vector form, mean over the replicates. Blue band: standard deviation of the Twin method over the replicates. Dashed purple: oracle with the fitted model, which knows the true phase of every cell and simulates the time point again with the fitted model. Dotted grey: the same oracle with the model that generated the data. The time point at 18 h has the fewest cells (116). **(C, D, E)** Spread of the single cell periods σ_T, obtained from the slope of a line fitted to the estimates of the 11 time points after the first one (the slope is not allowed to be negative), for the ramp (C) and the constant profile (D) in simulation, and for the real data (E). Same colors as in A. The two oracles have open markers (fitted model in purple, true generator in grey). Black bar: σ_T from the line fitted to the true values, mean over the 10 replicates (it is not zero for the constant profile, because the true values vary a little around 2.24 h). Small dots: the 10 replicates for the simulations, and 200 of the 2000 draws of a bootstrap over the cells for the real data (cells resampled within each time point, technical part held fixed, the same draws for both methods, the model not fitted again). Large markers: mean and standard deviation over the replicates (C, D) or over all 2000 draws (E). Cross: σ_T of the real estimates without the bootstrap.

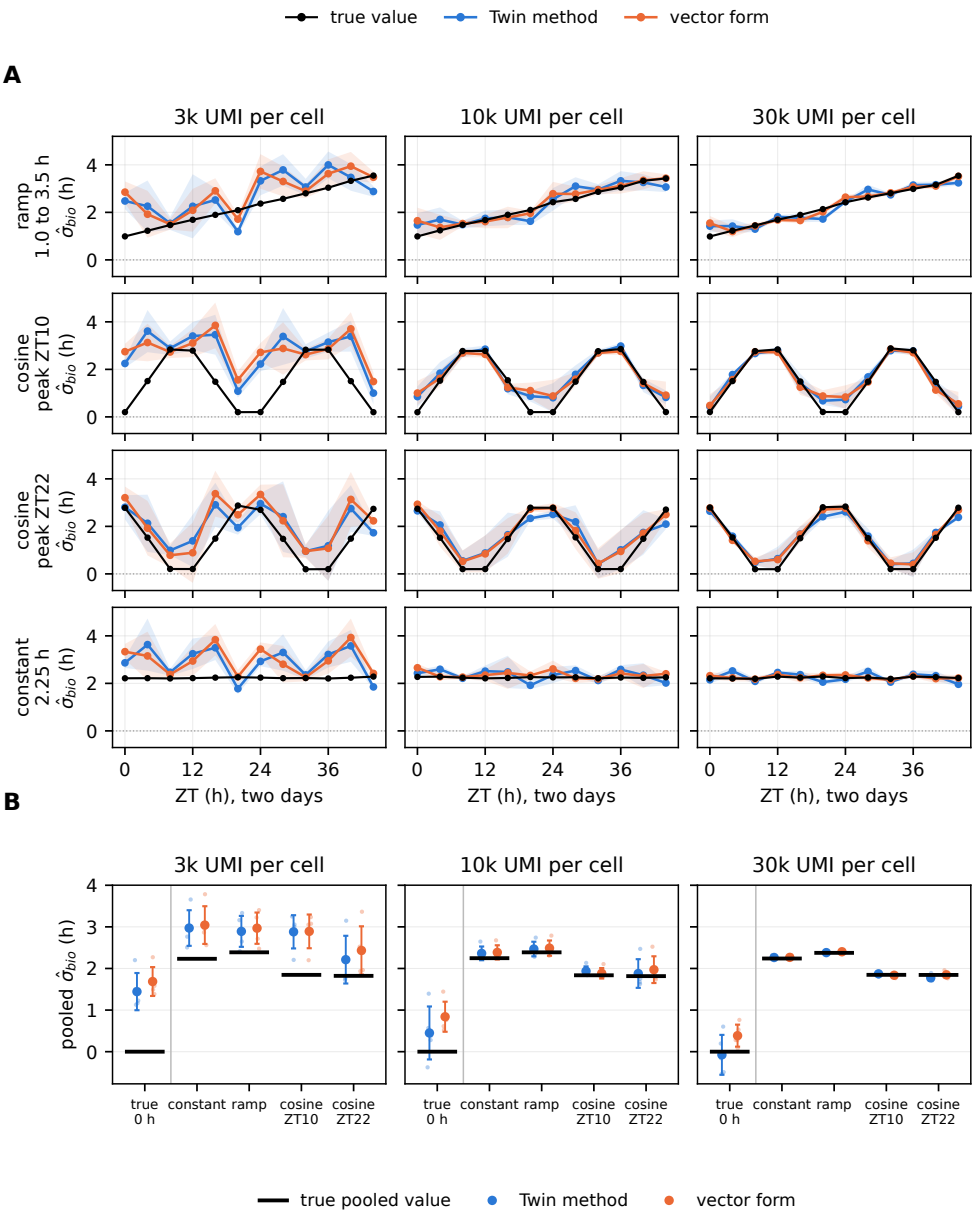


Figure Reb 4: **Simulations with a σ_{bio} that changes across time points.** 15 clock genes, 12 time points over two days, 500 cells per time point, 5 replicates per setting. Columns are the depths. **(A)** Estimated σ_{bio} at each time point for four true profiles (rows). Black: true value. Lines: mean over the replicates. Bands: standard deviation over the replicates. The true cosine is sampled every 4 h (0.2 to 2.8 h) and repeats on the second day. **(B)** The single value over all time points (Eq. 16 of the main text) for the same profiles (cosine ZT10 and ZT22 are the two cosines of A) and for a control with a true σ_{bio} of 0. Black bar: true value, the square root of the average of the true σ^2_{bio} weighted by cell number. Large markers: mean and standard deviation over the replicates. Small dots: the 5 replicates. Twin method and vector form are defined in the text and in the response to R2.1.

R2.3d

The equations for the aggregation and their corresponding description, both in the main text and response, are quite confusing because key quantities have been vaguely defined. What exactly do the words "condition" and "batch" mean here? What does the term σ_b refer to in Eq. 16 of the revised main text? None of these have been defined.

Response.

R2.4

The authors also comment that the aggregation procedure creates a requirement of sufficient ZT samples across 24 hours. In their simulations, they use 12 and 8 ZT samples, which is greater than the number of samples one would get from typical scRNA-seq datasets. For example, the Auerbach et al paper which the authors analyze, had only 4 samples spaced 6 hours apart. To better understand the limits of the aggregation procedure, the authors should demonstrate what happens in such limited datasets — would accurate estimates of a population σ_{bio} still be possible if there are just 4 samples?

Response.

R2.5

In the application of the method to phase inference in asynchronized FLASH-seq data, I appreciate the new results presented and agree that they provide more evidence that biologically meaningful oscillations are being reconstructed. However, I'm still puzzled by the fact that the mRNA and protein oscillations look identical, and not phase shifted at all. The protein oscillation peak should be delayed by 2–4 hours as per a vast body of prior literature, which is not reflected in the results here. How does one interpret this? Does this suggest that in fact there are significant errors being made in phase assignment to the single cells? If so, I would recommend discussing this in the Main Text and also toning down the statement "The inferred phase accurately reconstructed the oscillation of both the endogenous (Rev-Erbα) mRNA level and the Rev-Erbα-Venus protein fluorescence", since the meaning of "accurately" cannot really be quantitatively established in this setting.

Response.

R2.6

In the main text the authors say "Across all tissues, cells with higher posterior uncertainty σ_u showed larger phase errors than cells with lower σ_u (Fig. S3C, E, F), consistent with the per-cell correlation between σ_u and phase error". In the Discussion they say "The posterior uncertainty σ_u consistently predicts inference quality across simulations and diverse biological datasets, enabling principled quality control without external time labels." However, in the light of the new quantification of correlations between σ_u and error (ranging from 0.08 to 0.35 in Fig. Reb 10A), these statements seem quite disproportionate. Even the highest of these correlation values at 0.35 is weak at best, and it is also unclear whether any of these would pass statistical tests of significance. The authors should therefore report these magnitudes along with results from statistical tests in the Main text, and statements such as those above should be tempered accordingly.

Response.

Appendix A: the vector form

This appendix describes the vector form estimator summarised in the response to R2.1. It uses the notation of R2.1: group b of n_b cells, true phase $\theta \sim p_b$, inferred phase $\hat{\theta}$, $\rho(\theta) = \mathbb{E}[e^{i\hat{\theta}} \mid \theta] = r(\theta)\, e^{\mathrm{i}m(\theta)}$, and the approximations A1 and A2 of the Twin method. In scRitmo it is `estimate_phase_desynchrony(sigma_tech_method="deconvolution", deconv_form="vector")`.

Starting point.

Instead of correcting the approximations A1 and A2 one by one, we start from the exact identity of R2.1, Eq. (2). Its left side is what the data give (estimated by the mean $\bar{z}_b$ of $e^{\mathrm{i}\hat{\theta}_c}$ over the cells, with $\bar{R}_b = |\bar{z}_b|$). On the right, $\rho(\theta)$ is the technical part, which depends only on the model and the library sizes, and p_b is the biological part, whose width is $\sigma_{\mathrm{bio},b}$. The identity needs no expansion, and no assumption that r is constant or $m(\theta) = \theta$.

Step 1, measure ρ on a grid.

At each of $N = 24$ phases $\phi_k = 2\pi(k-1)/N$ (1 h apart) we simulate Twin cells (1000 per run, 5 runs pooled, so $M = 5000$), infer their phases and take the mean unit vector $\hat{\rho}_k = \frac{1}{M}\sum e^{\mathrm{i}\hat{\theta}}$ over the M cells at ϕ_k. This is a plain mean, so $\mathbb{E}[\hat{\rho}_k] = \rho(\phi_k)$. We keep the complex value, because its direction holds the bias.

Step 2, Fourier series.

Since ρ is periodic,

$$\rho(\theta) = \sum_{j=-J}^{J} c_j\, e^{\mathrm{i}j\theta}, \qquad c_j = \frac{1}{N}\sum_{k=1}^{N} \hat{\rho}_k\, e^{-\mathrm{i}j\phi_k}, \qquad J = \lfloor N/2 \rfloor, \tag{3}$$

where the term at $|j| = N/2$ is split in two halves, so that the series passes through every grid point.

Step 3, average over the biological distribution.

We assume p_b is a wrapped normal with center μ_b and width σ in radians, where σ is the trial value of $\sigma_{\mathrm{bio},b}$. For harmonic j (period $24/j$ h),

$$\mathbb{E}_{\theta\sim p_b}\left[e^{\mathrm{i}j\theta}\right] = \kappa_j(\sigma)\, e^{\mathrm{i}j\mu_b}, \qquad \kappa_j(\sigma) = e^{-j^2\sigma^2/2}. \tag{4}$$

The factor κ_j lies between 0 and 1. A distribution that is wide compared with the period covers both the peak and the trough of the harmonic, and the average shrinks toward zero. Putting Eq. (3) into the right side of Eq. (2) gives the expected mean vector of a group,

$$\bar{\rho}_b(\sigma) = \mathbb{E}_{\theta\sim p_b}\left[\rho(\theta)\right] = \sum_{j=-J}^{J} c_j\, \kappa_{|j|}(\sigma)\, e^{\mathrm{i}j\mu_b}. \tag{5}$$

The grid gives all the c_j, so this is a known function of σ. Under the conditions where Eq. 14 is exact (r constant, $m(\theta) = \theta$, and many cells), it reduces to Eq. 14.

Step 4, one equation with one unknown.

For independent cells, $\mathbb{E}[\bar{R}_b^2] = |\bar{\rho}_b|^2 + (1 - |\bar{\rho}_b|^2)/n_b$, where the second term comes from the n_b products of a cell with itself. The estimate $\hat{\sigma}_{\mathrm{bio},b}$ is the value of σ that solves

$$|\bar{\rho}_b(\sigma)|^2 + \frac{1 - |\bar{\rho}_b(\sigma)|^2}{n_b} = \bar{R}_b^2. \tag{6}$$

Here $\bar{R}_b$ and n_b come from the data, the c_j from the grid, and μ_b is the ZT of the group in the phase frame of the fitted model, so only σ is unknown. We search σ over $[0, \pi]$ and accept a root only if it is unique. If the data are tighter than the prediction at $\sigma = 0$ we set the estimate to zero. If no unique root exists, the group has no estimate. The estimand is $\sigma_{\text{bio},b}$, the spread of θ, and not B_b.

The group phase when the ZT is not known.

The steps above need μ_b. When the external time does not give the clock phase of the cells, as in the SABER-FISH data (time after dexamethasone), μ_b has to come from the data. The circular mean of the inferred phases is not a good choice, because the bias pulls it toward the trough of the aggregate circadian counts. The grid already measures this pull: $m(\theta) = \arg \rho(\theta)$ is the mean direction of the inferred phases of a synchronized group at true phase θ. We therefore take as μ_b the phase whose synchronized group has the observed mean direction, that is the solution of $\arg \rho(\mu) = \arg \bar{z}_b$ (the root closest to $\arg \bar{z}_b$). This undoes the shift of the mean by the bias. Where $m(\theta)$ is too flat to be inverted, which can happen at very low depth, we keep the circular mean. We also tried to solve for μ_b and σ together, matching both the direction and the length of $\bar{\rho}_b(\sigma)$ to $\bar{z}_b$, and it was not better.

We tested this on the simulation of R2.1, with 6 groups of 500 cells and 3 replicates, and one fit and one grid shared by all variants. Compared with the circular mean, the corrected phase lowers the mean absolute error of $\hat{\sigma}_{\text{bio},b}$ at 3k UMI per cell from 0.9 to 1.2 h to 0.4 to 0.8 h. This is the same as with the known ZT (0.3 to 0.8 h). At 10k it is between 0.05 h worse and 0.08 h better than with the known ZT, and at 30k the three choices of μ_b differ by 0.03 h or less. In scRitmo this is the option `debias_mean=True`, together with `use_circular_mean=True`.

What is removed.

A1 disappears because Eq. (5) averages ρ over the whole distribution p_b. A2 disappears because ρ contains the direction $m(\theta)$, so the bias is inside the prediction. The circular correction disappears because Eq. (6) compares mean vectors and does not add variances, and the finite n_b term is included. What remains is an offset between the phase origin of the fitted model and ZT 0 (0.2 to 0.55 h in our simulation), which affects both methods.

One σ_{bio} shared by all groups.

When all groups have the same σ_{bio}, Eq. 16 of the main text averages the solutions of Eq. (6) over the groups. We can instead average the equations and solve once. The step follows from the law of total expectation, applied twice. Inside group b, conditioning on θ gives Eq. (2), and from it $\mathbb{E}[\bar{R}_b^2] = L_b(\sigma_{\text{bio}})$, with $L_b(\sigma) = |\bar{\rho}_b(\sigma)|^2 + (1 - |\bar{\rho}_b(\sigma)|^2)/n_b$ the left side of Eq. (6). Across groups, take a cell at random, so that its group b has probability $w_b = n_b / \sum_{b'} n_{b'}$, and condition on b. Then $\mathbb{E}_b\big[\mathbb{E}[\bar{R}_b^2 \mid b]\big] = \sum_b w_b \, L_b(\sigma_{\text{bio}})$, which is again exact. Matching it to the data gives one equation in σ,

$$\sum_b n_b \, \bar{R}_b^2 = \sum_b n_b \, L_b(\sigma), \tag{7}$$

which we solve as Eq. (6), with an estimate of zero when the data of all groups together are tighter than the prediction at $\sigma = 0$. Each L_b keeps its own μ_b, so the technical part still changes from group to group. Only σ is shared. Two things improve over Eq. 16 of the main text. The noise of the groups cancels before the solve, which is not linear in $\bar{R}_b^2$. And the estimate is set to zero at most once per data set, not once per group, which is what makes Eq. 16 of the main text too large when the true σ_{bio} is small.

In the simulation of R2.1 with 10 replicates, the error of the single value at a true spread of zero

is 0.25, 0.11 and 0.09 h at 3k, 10k and 30k, against 0.50, 0.39 and 0.32 h for Eq. 16 of the main text with the vector form. At a spread of 2.5 h and 3k it is 0.17 against 0.26 h. In the other settings the two differ by 0.01 h or less. The result is the same when we remove 1 or 2 of the 6 groups. When $\sigma_{\mathrm{bio},b}$ changes across groups, Eq. (7) gives approximately the average of $\sigma^2_{\mathrm{bio},b}$ weighted by $n_b|\bar{\rho}_b|^2$ and not by n_b, so there we keep the estimates per group, as for SABER-FISH. In scRitmo it is `solve_vector_shared`.

Assumptions that remain.

The wrapped normal shape of p_b; a fitted model close to the data generating process; one library size distribution shared by all groups (otherwise we build one grid per group); independent cells; and a grid large enough that its own sampling noise is small next to that of the data. The case where $\sigma_{\mathrm{bio},b}$ changes from group to group, as in the SABER-FISH data, is tested in R2.3b and R2.3c.

References